

Reward Design under Shared Connectivity Constraints for Cellular UAV Control

Guillem Moreno Garcia and Evgenii Vinogradov
Universitat Politècnica de Catalunya, Barcelona, Spain
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Abstract—Reward comparisons for cellular UAV control must preserve the communication requirements that make a mission successful. We compare a master’s thesis reward with a replacement within the same strict sampled connectivity system, V2. The original reward arm, V1.5, retains reciprocal radio utilities, a binary approach bonus and source penalties; it shares V2’s dynamics, observations, actions, residual navigation prior, safety filter, PPO settings and termination. Five training seeds per learned arm and 400 fresh shared routes support the comparison. Full V2 achieves 95.7% standard and 87.3% longer joint success, versus 95.7% and 85.8% for V1.5. The primary standard difference is +0.0 percentage points, with a 95% crossed seed and route bootstrap interval $[-2.8, 2.7]$. The primary comparison does not establish improved completion from reward replacement. An arrival reward control and four deterministic references expose communication tradeoffs and the shared controller’s role. Every successful mission meets the sampled RSS and buffer constraints. These results concern a declared surrogate, not the unavailable original simulator or physical connectivity between samples.

Index Terms—Cellular UAV, connectivity constraints, reward design, handover, trajectory control, resource allocation, reinforcement learning, reproducibility.

I. INTRODUCTION

A master’s thesis studies joint trajectory, cell association and resource control using Barcelona ray tracing data [1, pp. 6–7, Sec. 3.1]. Its objective weights delay, interference and handovers [1, p. 10, Eq. (11)], with minimum serving RSS and bounded queue constraints [1, p. 11, Eq. (12), C2 and C4]. The thesis reports wandering and prioritization of connectivity over destination completion [1, p. 19, Sec. 7.2; p. 21, Sec. 8]. Positive reciprocal radio utilities and a fixed approach bonus [1, p. 12, Eqs. (13)–(14)] motivate an empirical reward audit, but do not establish the cause of behavior in unavailable original code.

The central question is whether replacing that reward improves joint mission success when every other component is the same V2 system. V1.5 denotes the written original reward on V2, created as a new controlled intervention. Both arms retain V2’s strict sampled connectivity rules, arrival termination, navigation prior and prospective safety filter. We do not restore the original continued training after arrival [1, p. 12, Sec. 5.2]. The experiment therefore measures the reward package conditional on V2, not the complete original agent or individual reward terms in isolation.

We contribute a transparent reward comparison, fresh matched tests, all training seeds, explicit failures and reproducible artifacts. PPO [2] is an established algorithm; we claim

no new learning method. Thesis citations give printed pages; add two for the PDF viewer page. The original simulator and trained policy remain unavailable, and exact source dataset version identity is unverified.

II. SOURCE REQUIREMENTS

A. Which Communication Requirements Apply?

We use the thesis requirements, not a newly imposed application service level. The RSS minimum is -96 dBm and the buffer is listed as 160 KB [1, p. 15, Table 3]; we retain the explicit 1,280,000 bit interpretation. The written C2 inequality allows equality and applies at every modeled time [1, p. 11, Eq. (12)]. Delay remains an optimized quantity, defined in the source as queue divided by service rate [1, p. 9, Eq. (4)]. Neither a two second packet deadline nor an empty queue arrival condition is introduced. A successful arrival can therefore retain queued data; it must not be described as completed data delivery.

The distinction between formulation and enforcement matters. The source reward penalizes RSS violations [1, p. 12, Eq. (14)], which does not establish that its unavailable implementation guaranteed C2. Its direct data dump penalty was removed during development [1, pp. 20–21, Sec. 7.3]. We compare verified written definitions with our executable rules, not inferred behavior of missing original code.

III. SHARED V2 SYSTEM

A. Shared Physical Assumptions

We preserve the received native resolution RSS map, selecting the first operator’s 87 stations. Matching source metadata includes 2.1 GHz and 100 m altitude [1, p. 7, Table 2; p. 15, Table 3]; this does not prove an identical original dataset version. The raw private HDF5 is not modified or published. The provisional -128 sentinel interpretation remains unverified.

The shared model retains a one second decision step, acceleration norm limit 5 m/s^2 , speed limit 25 m/s , 200 s deadline, and a 100 kJ budget under the engineering energy proxy $100 + 0.4v^2$ watts. These are shared V2 settings. They differ from the source’s 0.1 s table step and 100 m/s speed setting [1, p. 15, Table 3]; the source also gives a conflicting 1 ms interval in prose [1, p. 14, Sec. 6.1].

The radio proxy retains 12 resource groups, half occupied by a deterministic station pattern. Interference sums linear received power from cochannel background stations. Capacity is $1.44 \cdot 10^6 \log_2(1 + \text{SINR})$ bit/s, with noise -103.41 dBm

and offered traffic 200,000 bit/s. The source's printed rate and SNR expressions differ [1, p. 10, Eqs. (7)–(8)], and its background occupancy is described as random [1, p. 14, Sec. 6.1]. We retain these declared surrogate assumptions to separate the current repair from a new propagation or traffic model. The interference is a downlink proxy, not a calibrated uplink quantity matching the source's terminology [1, p. 10, Eq. (9)].

The shared motion preserves inertial velocity and uses trapezoidal position integration, unlike the source scalar speed/direction update [1, p. 9, Eq. (2)]. The queue receives constant traffic and executed handover overhead, subtracts this step's service, then records overflow before clipping. Delay is postservice backlog divided by a rate floor of one bit/s, censored at 200 s. The source supplies the queue/rate concept and packet formulation [1, pp. 9–10, Eqs. (4)–(6)]; our constant bit rate and service proxy do not reproduce individual packet latency.

A full grid scan finds at least 38 selected operator stations above -96 dBm at every location. Direct trajectories can therefore be appropriate; actual association restrictions and interference still matter. Curvature is not an objective. Buildings affect the inherited propagation map, but no obstacle collision model is implemented.

B. Joint Success and Failure

Let d_t denote destination distance and q_t the queue after the sampled arrival/service update. Joint success requires arrival by the deadline with $d_T \leq 10$ m and $\|v_T\| \leq 2$ m/s, together with

$$\text{RSS}_{k_t}(p_t) \geq -96 \text{ dBm}, \quad 0 \leq q_t \leq 1,280,000 \text{ bits}, \quad \forall t. \quad (1)$$

The 10 m tolerance follows the source table [1, p. 15, Table 3]; the numeric stopping tolerance is our explicit interpretation of stopping at the destination [1, p. 9, Sec. 3.2]. Map and energy conditions also apply. A single sampled RSS violation or any buffer overflow ends a failed mission. Failure takes precedence over simultaneous arrival. Initial RSS is checked; queue clipping for storage cannot erase an overflow flag.

This enforces the constraints at modeled samples. It is not a guarantee between the one second samples or within the queue update. No minimum SINR, packet latency threshold, or final queue draining requirement is claimed.

C. Actions, Residual Motion and Safety Filter

The source includes BS and RBG actions [1, p. 12, Sec. 5.2]. We expose 61 categorical options: stay, or five candidate station slots by 12 RBGs. Candidates are serving plus the four strongest RSS stations. Masks reject occupied groups and duplicated serving station slots. A new station must have current RSS more than 3 dB above the serving RSS and meet the minimum, following the source's A3 direction [1, p. 9, Eq. (3); p. 15, Table 3]. Network decisions occur each modeled second. Resource changes within the same station are allowed and counted separately from handovers. Handovers generate

4,800 control bits [1, p. 15, Table 3]; switching interruption is not modeled.

The observation has 156 features covering geometry, velocity, time, queue, energy, serving SINR, candidate RSS and positions, option capacities and masks. Two Gaussian outputs define bounded residual acceleration commands. We project a destination directed braking command onto the unit disk, add half of the tanh transformed residual, and apply the shared acceleration projection. All learned arms use this prior; they do not learn basic flight entirely from scratch. The source's fully discrete action and smaller state differ [1, p. 12, Sec. 5.2].

A shared safety filter predicts the proposed next position, RSS and queue. It first tries another admissible station/resource pair for the same motion. If none is feasible, it invokes the local motion/network search. If no candidate repairs the step, the flight fails normally. The filter has prospective access to the known map and is an explicit control component, not learned competence. Interventions are recorded. It cannot establish global reachability.

D. The V2 Full and Arrival Rewards

Using source equal weights and scales [1, p. 15, Table 3], define

$$C_t = 0.35 \frac{10D_t}{1 + 10D_t} + 0.30 \frac{10^5 I_t}{1 + 10^5 I_t} + 0.35 \frac{100H_t}{1 + 100H_t}. \quad (2)$$

Here D_t is the backlog/service delay proxy in seconds, I_t interference in watts, and H_t a handover indicator. These are one minus the source reciprocal utilities [1, p. 12, Eq. (13)], removing the positive reward for merely remaining active while retaining all communication components. The reciprocal cost still saturates smoothly, which can favor substantial delay in exchange for fewer handovers; it is not a latency guarantee.

The V2 full and arrival arms use

$$\Phi_t = -\frac{d_t + \|v_t\|^2 / (2a_{\max})}{100 \text{ m}}, \quad (3)$$

with terminal potential zero. The base reward is $0.99\Phi_{t+1} - \Phi_t - 0.05$, plus 20 for joint success and minus 20 for terminal failure. The arrival arm uses this base alone; the full arm subtracts C_t . Both have identical hard failure rules and control interfaces. This is not a comparison to the original trained thesis policy or a claim that a finite scalar reward proves optimal lexicographic control.

IV. ORIGINAL REWARD INTERVENTION AND EXPERIMENTAL PROTOCOL

A. VI.5: Original Written Reward on V2

For executed postfilter delay D , interference I and handover h , use

$$r_{\text{orig}} = \frac{0.35}{1 + 10D} + \frac{0.30}{1 + 10^5 I} + \frac{0.35}{1 + 100h} + 12\mathbb{I}[d_{t+1} < d_t] - 10b - 10o - 10s. \quad (4)$$

The source supplies the algebra, equal weights and coefficients [1, p. 12, Eqs. (13)–(14); p. 15, Table 3]. Here b is a boundary violation, o is $\text{RSS} \leq -96$ dBm, and s is speed ≥ 100 m/s. Energy exhaustion replaces the whole reward with -10 [1, p. 12, Sec. 5.2]. No new arrival, timeout or overflow payment is added. The source’s removed energy utility and data dump penalty remain absent [1, pp. 20–21, Sec. 7.3].

The reward retains its RSS equality penalty although the joint constraint permits equality. Its speed threshold remains 100 m/s, unreachable under the V2 25 m/s cap. Substituting 25 m/s would invent a penalty on routine cruising. The same V2 transition supplies radio inputs, physics, filter interventions and terminal outcomes to all arms. The wrapper changes only returned reward and accumulated raw return. Full reward replacement changes the entire package; the effects of shaping, terminal payments and removing positive per step utilities are not separately identified.

B. Shared Training and Fixed Checkpoints

PPO uses separate two layer 64 unit tanh actor and critic networks, 64 lanes, 128 step rollouts, four epochs, minibatches of 512, learning rate 3×10^{-4} , Adam epsilon 10^{-5} , discount 0.99, GAE 0.95, clip 0.2, entropy weight 0.005, value weight 0.5 and gradient norm limit 0.5. Two CPU threads are used. Running discounted return variance normalizes rewards, clipped to $[-10, 10]$. This shared processing responds to changed raw rewards; potential algebra alone does not establish invariance after these transforms. The source uses Stable Baselines PPO [1, p. 13, Sec. 5.3] and different training settings [1, pp. 15–16, Sec. 6.2 and Table 4].

All arms use the exact 4,096 V2 training routes spanning 200–1800 m, 64 standard validation routes and 32 longer validation routes. Five initialization seeds, 2101–2105, and 524,288 interactions per policy are shared. Five original reward policies are newly trained from scratch: 2,621,440 new interactions. All ten completed full and arrival V2 policies are reused; they used identical training data, seeds and budget. Their hashes were fixed before new training and test evaluation. No seed is selected for its performance. Different policies can reset at different times, so equal seeds and budgets do not imply identical realized transitions. Every run uses its final budget checkpoint.

Earlier V2 test outcomes were known. This protocol was frozen before the new training and fresh test evaluation; it is not independently registered or claimed to predate the earlier V2 study. No new hyperparameter tuning or policy selection uses the fresh tests.

C. Fresh Tests and Prespecified Analysis

Generate 200 standard routes of 200–1000 m using seed 53012 and 200 longer routes of 1000–1800 m using seed 53013. Exact route pairs are disjoint from all prior saved routes; every controller shares initial conditions and load phases. Both lengths occur in training. This is route holdout on one map, not geographic generalization or length extrapo-

TABLE I
STRICT JOINT SUCCESS ON THE NEW SHARED ROUTES (PERCENT).

Controller	Standard	Longer	Per split
PPO original (V1.5)	95.7	85.8	1,000
PPO full (V2)	95.7	87.3	1,000
PPO arrival (V2)	94.9	85.1	1,000
Goal RSS	94.0	84.5	200
Goal radio	96.5	88.5	200
Joint one step	96.0	90.5	200
Joint three steps	92.0	90.5	200

lation. The source does not document equivalent independent splits [1, pp. 15–16, Secs. 6.2–6.3].

The primary contrast is full minus original standard joint success. Use 5,000 crossed seed and route bootstrap draws, seed 63000, to estimate a 95% interval. An interval entirely above zero supports improved primary completion under this design. Longer completion is prespecified secondary. Five seeds on 200 shared routes are not 1,000 independent training replications. Secondary intervals are descriptive and unadjusted for multiple comparisons.

Secondary metrics use the common successful seed/route pairs, with their count visible. Unconditional success and failure rates remain separate. No percentage is computed for logarithmic SINR. Three learned arms supply 6,000 episodes; four deterministic references supply 1,600, totaling 7,600 new final evaluations. These records are separate from the previously evaluated V2 routes.

D. Deterministic References and Verification

Goal RSS uses nominal braking motion and strongest admissible association. Goal radio uses the same motion and prospective normalized radio cost selection. One step and three step joint search examine nine initial motion primitives, varying direction and speed. The three step version follows each initial choice with two predicted reference steps. All share the same safety filter, which can alter nominally straight paths. These local methods can fail and are not a reproduction of the source greedy controller [1, pp. 13–14, Sec. 5.4].

All 73 tests pass, including unchanged states, observations, masks and terminal records under identical actions across reward arms. Failure checks cover RSS, overflow, boundary, energy and timeout as well as arrival. A syntax comparison verifies that copied PPO differs only in its environment import and module description. Small full reward training reproduces frozen V2 parameters exactly; original reward training repeats exactly in its deterministic check. All 7,600 new final episode records replayed exactly. Source, dataset and checkpoint hashes remain unchanged. These checks verify software reproduction, not physical model validity.

V. RESULTS ON FRESH SHARED ROUTES

A. Primary Reward Contrast

Full achieves 95.7% standard success versus 95.7% for V1.5. The paired difference is +0.0 percentage points, 95% in-

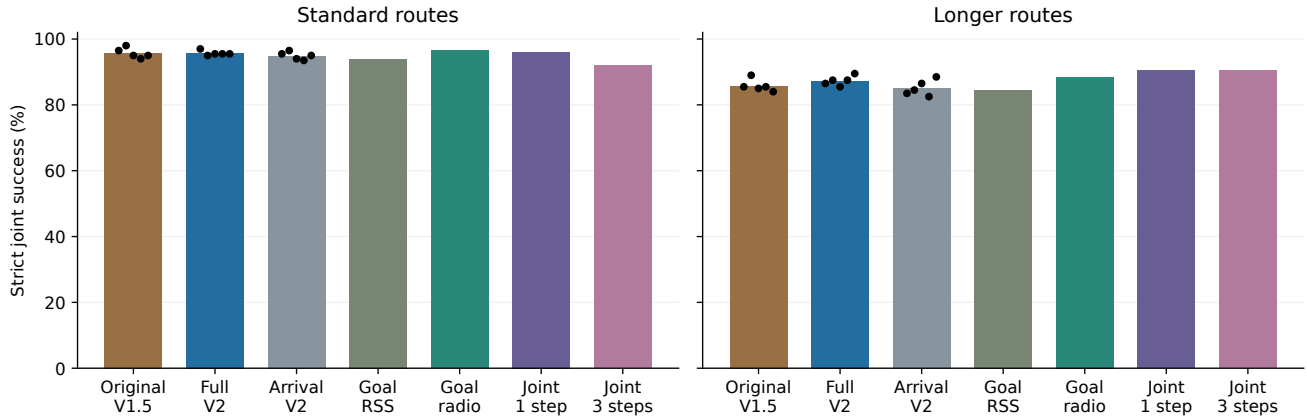


Fig. 1. Joint success on fresh shared routes. Bars aggregate episodes; dots show individual learned training seeds, not confidence intervals. Deterministic controllers have one evaluation per route.

TABLE II
FIRST FAILURES: RSS / BUFFER / TIMEOUT COUNTS. NO ENERGY OR
BOUNDARY FAILURES OCCURRED.

Controller	Standard	Longer
PPO original (V1.5)	40 / 3 / 0	136 / 6 / 0
PPO full (V2)	34 / 9 / 0	112 / 15 / 0
PPO arrival (V2)	51 / 0 / 0	147 / 0 / 2
Goal RSS	12 / 0 / 0	31 / 0 / 0
Goal radio	7 / 0 / 0	21 / 2 / 0
Joint one step	8 / 0 / 0	18 / 0 / 1
Joint three steps	4 / 0 / 12	11 / 0 / 8

terval $[-2.8, 2.7]$. The primary comparison does not establish improved completion from reward replacement. An interval containing zero is not proof of equivalence. On longer routes the difference is +1.5 points, interval $[-2.9, 6.1]$. The longer route interval also includes zero; its observed difference does not establish improvement. Every training seed appears in Table IV.

B. Failures and Communication Tradeoffs

Table II reports first failure reasons. Episodes stop on the first violation; hypothetical later failures are unknown. Every successful flight has zero sampled RSS outage and zero dropped bits.

Full versus original secondary comparisons contain 925 common successful standard seed/route pairs and 781 longer pairs. Table III reports conditional paired values and intervals. On standard common successes, accumulated radio cost changes by -25.2%, while delay changes from 1.94 to 5.83 s. Flight time changes by -3.14 s and handovers by -6.40. On longer common successes, delay changes from 2.86 to 6.82 s. Interference differences remain inconclusive on both splits. A favorable aggregate cost is not interchangeable with delay, interference, handovers or completion. Different successful subsets must not be treated as matched counterfactual outcomes.

C. Other Controls

Full versus arrival completion differences are +0.8 points $[-1.8, 3.4]$ on standard routes and +2.2 points $[-2.3, 6.7]$ on longer routes. Both arms enforce strict constraints. Goal radio attains 96.5% and 88.5% joint success. Joint search has the highest observed longer completion on this route set, while Goal radio has the highest standard completion. These observed rankings are not a demonstrated overall superiority claim. These references expose performance available from the shared model rather than demonstrating learned superiority.

D. Illustrative Long Route

Figures 2–3 show route 53013_0023, the longest Euclidean start to goal distance among all 200 declared longer test routes (1,784.7 m). Ties use scenario ID; learned seed 2101 is retained. Selection uses geometry only. Dense replay of the three original 200 route batches matched all 600 records exactly; this illustration adds no independent observations or training.

All three selected flights succeed: original reward takes 85 s and 18 handovers, full takes 78 s and one handover, and Goal radio takes 77 s and two handovers. Mean delay proxies are respectively 0.78, 2.79 and 0.87 s. Full's queue approaches capacity while RSS stays above the minimum. Paths largely overlap under the shared navigation aid, despite different service behavior. Full's accumulated radio cost is higher than original on this route, unlike the average contrast. Aggregate results determine the conclusions.

VI. INTERPRETATION AND LIMITATIONS

The contrast isolates the reward package conditional on V2. It does not identify individual terms or behavior without the navigation prior and prospective filter. These components may compensate for reward deficiencies. Removing them would require a separate experiment and fresh evaluation.

Original speed limits, motion, discrete action parameterization, continued training after arrival and unknown implementation details differ. Reconstructing the reward on V2

TABLE III
FULL MINUS ORIGINAL ON 925 STANDARD AND 781 LONGER COMMON SUCCESSFUL PAIRS. SECONDARY INTERVALS ARE DESCRIPTIVE AND UNADJUSTED.

Split	Metric	Original	Full	Difference	95% interval
Standard	Flight time (s)	31.252	28.112	-3.139	$[-3.509, -2.808]$
Standard	Accumulated radio cost	7.094	5.309	-1.785	$[-2.194, -1.389]$
Standard	Delay proxy (s)	1.942	5.827	+3.886	$[2.731, 5.066]$
Standard	Handovers	6.843	0.443	-6.400	$[-8.501, -4.366]$
Standard	Interference (μW)	0.753	0.748	-0.004	$[-0.017, 0.009]$
Standard	Energy proxy (kJ)	7.651	7.788	+0.137	$[0.115, 0.157]$
Standard	SINR (dB)	-8.322	-9.305	-0.983	$[-1.483, -0.488]$
Standard	Path length (m)	547.596	551.204	+3.608	$[2.947, 4.273]$
Standard	Network overrides	0.107	0.478	+0.371	$[0.243, 0.504]$
Standard	Motion overrides	0.008	0.016	+0.009	$[-0.004, 0.028]$
Longer	Flight time (s)	65.394	60.145	-5.250	$[-5.815, -4.804]$
Longer	Accumulated radio cost	16.800	13.803	-2.997	$[-3.765, -2.257]$
Longer	Delay proxy (s)	2.857	6.818	+3.961	$[2.657, 5.145]$
Longer	Handovers	15.826	1.528	-14.298	$[-19.543, -9.101]$
Longer	Interference (μW)	0.794	0.791	-0.003	$[-0.015, 0.009]$
Longer	Energy proxy (kJ)	18.523	18.957	+0.434	$[0.394, 0.476]$
Longer	SINR (dB)	-8.773	-9.957	-1.184	$[-1.652, -0.712]$
Longer	Path length (m)	1345.958	1349.568	+3.610	$[2.896, 4.321]$
Longer	Network overrides	0.467	1.757	+1.289	$[0.882, 1.637]$
Longer	Motion overrides	0.024	0.044	+0.019	$[-0.010, 0.062]$

Longest declared test route: 1,784.7 m | Same start, goal and traffic

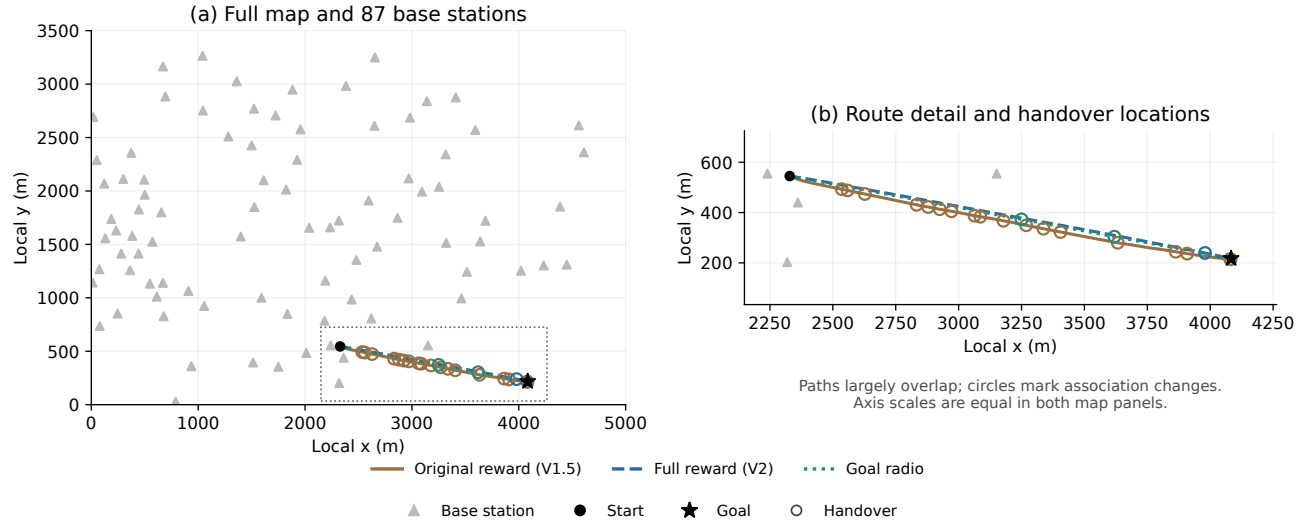


Fig. 2. Longest declared longer test route, with the full station map and route detail at equal coordinate scales. Triangles are the 87 Operator 1 base stations; colored circles mark executed handovers. Original reward, full reward and Goal radio share start, goal and traffic. All three succeed. Nearly overlapping paths do not imply identical communication service.

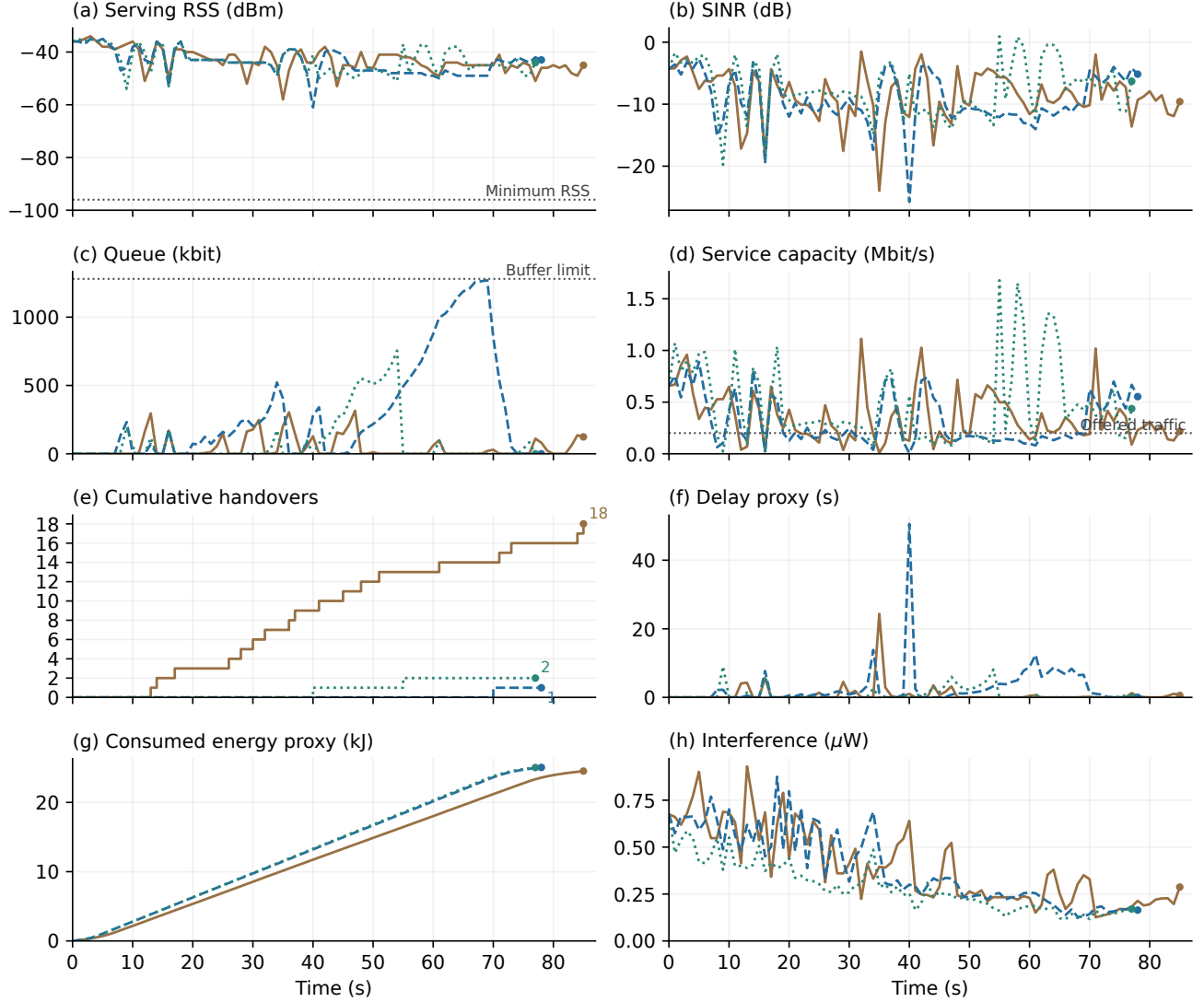
cannot identify the cause of the thesis's wandering [1, p. 21, Sec. 8]. Exact dataset version and sentinel meaning remain unverified. One map, static load, ideal service and uncalibrated energy limit inference. No physical continuity, packet latency, collision safety or deployment claim follows. A successful arrival can still retain queued data. Every table here uses the newly declared route seeds; earlier V2 results remain separate.

VII. CONCLUSION

The primary comparison does not establish improved completion from reward replacement. The full reward trades fewer handovers and shorter flights for higher delay, slightly higher proxy energy and lower SINR on common successful routes. The experiment compares the written original reward and a replacement under identical sampled constraints and control machinery, retaining failed missions and communication tradeoffs. Further claims require independent maps, traffic and controller ablations rather than tuning on these test routes.

Communication and energy along the same 1,784.7 m route

— Original reward (V1.5) — Full reward (V2) ... Goal radio



Initial and executed samples at 1 s intervals; each curve stops at its episode endpoint.

Fig. 3. Communication and energy on the same route and controller runs as Fig. 2. Curves include the initial state and each executed one second sample, stopping at their own episode endpoints. Dotted horizontal lines show minimum RSS, buffer capacity and offered traffic. Capacity is the model's potential service rate, not measured delivered throughput; delay is the queue divided by capacity proxy. Energy is the declared engineering proxy. No sampled outage or buffer overflow occurs; successful original reward arrival retains 123.65 kbit, which the shared requirements permit. Connecting sampled points does not establish connectivity between samples.

APPENDIX A ARTIFACTS AND REPRODUCTION

The repository includes all 15 final checkpoints, source, configuration, raw records, analysis, source snapshots and this paper: https://github.com/unworthyzeus/UAV_joint_handover_trajectory_optimization. Ten are unchanged V2 policies and five are new V1.5 policies. The private dataset belongs at `dataset/Barcelona_dataset_January.h5`; the setup guide records its checksum, dependency versions, exact commands and outputs. Note 36 is the frozen protocol, note 37 records results, and note 38 records delivery validation. The README documents the exact V2 differences from the TFM with page locators. This is a research draft, not an accepted publication.

APPENDIX B EVERY SEED AND OBSERVATION LAYOUT

TABLE IV
EVERY LEARNED SEED: JOINT SUCCESS PERCENTAGES.

Arm	Seed	Standard	Longer
PPO original (V1.5)	2101	96.5	85.5
PPO original (V1.5)	2102	98.0	89.0
PPO original (V1.5)	2103	95.0	85.0
PPO original (V1.5)	2104	94.0	85.5
PPO original (V1.5)	2105	95.0	84.0
PPO full (V2)	2101	97.0	86.5
PPO full (V2)	2102	95.0	87.5
PPO full (V2)	2103	95.5	85.5
PPO full (V2)	2104	95.5	87.5
PPO full (V2)	2105	95.5	89.5
PPO arrival (V2)	2101	95.5	83.5
PPO arrival (V2)	2102	96.5	84.5
PPO arrival (V2)	2103	94.0	86.5
PPO arrival (V2)	2104	93.5	82.5
PPO arrival (V2)	2105	95.0	88.5

TABLE V
IDENTICAL V2 OBSERVATION LAYOUT IN ALL THREE LEARNED ARMS.

Indices	Meaning
0–1	Position / map extent
2–3	Unit goal direction
4	Goal distance / 1,000 m
5–6	Radial/lateral velocity / 25 m/s
7–8	Speed / 25; elapsed steps / 200
9–10	Queue and consumed energy fractions
11–13	Load phase, RBG and serving SINR
14–18	Candidate RSS values
19–28	Candidate relative coordinates
29–33	Candidate station indices
34–94	Log capacity for 61 options
95–155	Network mask

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